Date:

Ex. No: 6 VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM

Aim:

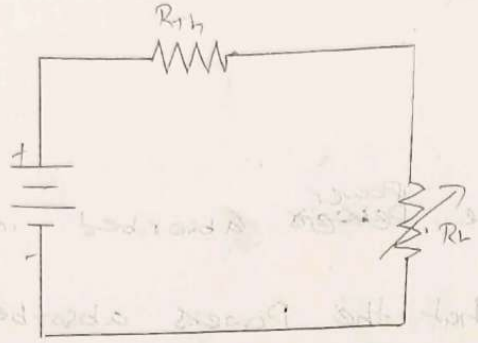
To measure the Power ~~Power~~ absorbed in a load and to verify that the Powers absorbed in a load is maximum only when load resistance is equal to the source resistance.

APPARATUS REQUIRED:-

S.No	name of the Apparatus	Range/Rating	Quantity
1.	Voltmeter	(0-15V) MC	1
2.	Ammeter	(0-500mA) MC	1
3.	Resistor	560 Ω 470 Ω	2 1
4.	RPS (DC SUPPLY)	15V	1

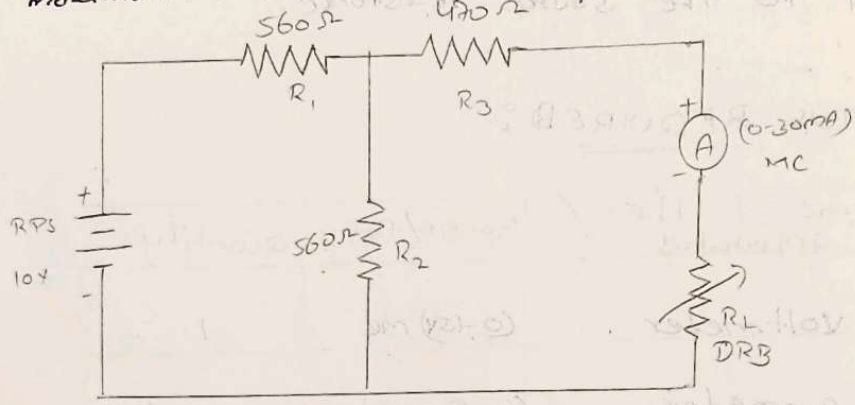
THEORY:-

The maximum Power transfer theorem state's that's "A load will recive maximum Power from a linear bilateral DC Network when it's total resistive value is exactly equal to the thevenin's resistance of the network as seen by load".

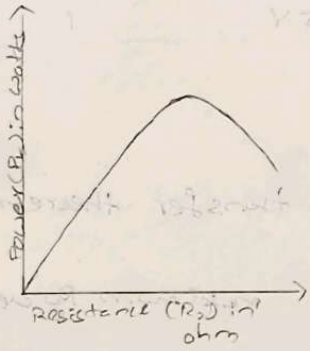


Circuit Diagram :-

For maximum power transfer theorem :-



Model Graph :-

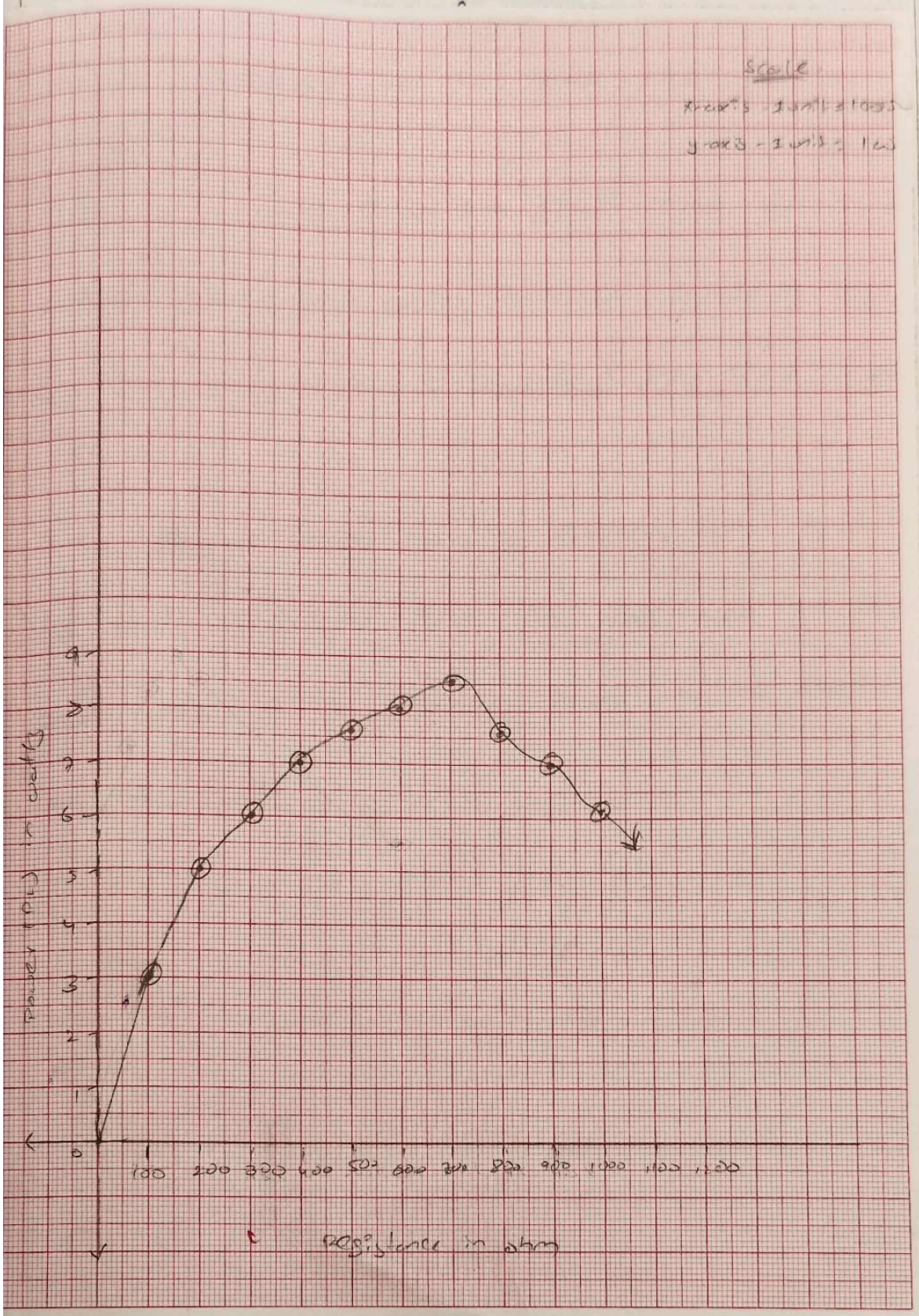


$$I_S = \frac{V}{R_S} = \frac{10}{25.5}$$

$$= 0.013A$$

$$I_N = 0.013 \times \frac{470}{470+560}$$

$$I_N = 6.75\ mA$$


Scale
x-axis - 1 unit = 100Ω
y-axis - 1 unit = 1W
Power (P) in watts
Resistance in ohm
0
1
2
3
4
5
6
7
8
9
100
200
300
400
500
600
700
800
900
1000
1100
1200

10. For the network in figure (a), maximum power will be delivered to the load when $R_L = R_{Th}$.

• In a simpler form the circuit may contain a voltage source (V_s) having internal resistance (R_s) and connected across a load (R_L). To Find Power.

The maximum power transfer theorem tells us that the load should be equal in magnitude to the source resistance for maximum power to be absorbed by the load.

TABLE (1): FOR THEORETICAL CALCULATION : $R_{th} = 750\Omega$

S.No	Load resistance (R_L) in ohm	Load current (I_L) in mA	Load voltage (V_L) in volts	load Power (P_L) in watts
1.	750Ω	3.33A	2.49 V	8.39 mW

Table (2): For Practical Calculation:

S.No	Load resistance (R_L) in ohm	load current (I_L) in amps	load voltage (V_L) in volts	load Power (P_L) in watts
1.	330	4.2×10^{-3}	1.551	7.29×10^{-3}
2.	470	4.20×10^{-3}	1.979	8.29×10^{-3}
3.	800	3.3×10^{-3}	2.64	8.71×10^{-3}
4.	1k	2.4×10^{-3}	2.9	8.41×10^{-3}
5.	2k	1.56×10^{-3}	3.22	6.42×10^{-3}

Theoretical calculation:-

$V_s = 10V$

By VDR $\Rightarrow V_{th} = V_s \times \frac{R_2}{R_2 + R_1}$

$V_{th} = 10 \times \frac{560}{1120}$

$V_{th} = 5V$

$R_{th} = R_L$ and R_2 is parallel

$= \frac{R_1 R_2}{R_1 + R_2} = \frac{560 \times 560}{560 + 560} = \frac{313600}{1120} = 280\Omega$

This is series with R_2:

$R_{eq} = 280 + 470 = 750\Omega$

PROCUDURE :-

1. Make connections as per the circuit diagram.

2. change the resistor R_L whose value close to R_{TH}, measure the corresponding V_L, I_L and calculate the power P_L and enter into the table (2).

3. Plot a graph b/w R_L and P_L and find the R_L corresponding to maximum-power transfer.

4) verify the measured values of R_L at maximum power transfer as same as calculated and found graphically.

S.NO	Description	marks alloted	marks obtained
1.	Experimental setup	20	20
2.	observation/Readings	20	18
3.	calculation/Results	20	18
4.	viva	20	18
5.	Record	20	18
6.	Total	100	92

RESULT :-

Thus, maximum power transfer theorem is verified practically.